

- Description

This module manages a smart tunnel air quality monitoring system, ensuring that:

1. **The concentration of harmful gases (CO, NO₂, PM2.5) is continuously measured and compared against safety limits.**
2. **If gas levels exceed the maximum safety threshold, an alert is triggered and ventilation is activated.**
3. **If the air quality improves, the ventilation system automatically reduces power to save energy.**
4. **A log of all air quality readings, alerts, and ventilation adjustments is maintained for environmental compliance.**

This system helps ensure safe air quality in tunnels, reduces pollution risks, and optimizes energy use in ventilation systems.

- Specification

module TunnelAirQualityMonitoring

var currentCO, maxCO, currentNO2, maxNO2, currentPM25, maxPM25: real

var ventilationActive: boolean

var log: sequence of string

process updateAirQuality(changeCO: real, changeNO2: real, changePM25: real)

functional scenario form:

currentPM25 <= maxPM25 && ventilationActive = false, log = log $\oplus$ ("Air Quality Normal. Ventilation Standby."), Air quality normal, no action needed.

||

currentPM25 > maxPM25 && ventilationActive = true, log = log $\oplus$ ("Air Quality Alert! Ventilation Activated."), Air quality alert, ventilation activated.

end process

end module

- Java Code

```
package code;
import java.util.*;
class TunnelAirQuality {
    private double maxCO;
    private double maxNO2;
    private double maxPM25;
    private double currentCO;
    private double currentNO2;
    private double currentPM25;
    private boolean ventilationActive;
    private List<String> log;
    public TunnelAirQuality(double maxCO, double maxNO2, double maxPM25
, double initialCO, double initialNO2, double initialPM25) {
        this.maxCO = maxCO;
```

```

        this.maxNO2 = maxNO2;
        this.maxPM25 = maxPM25;
        this.currentCO = initialCO;
        this.currentNO2 = initialNO2;
        this.currentPM25 = initialPM25;
        this.ventilationActive = false;
        this.log = new ArrayList<>();
    }

    public void updateAirQuality(double changeCO, double changeNO2, double changePM25) {
        currentCO += changeCO;
        currentNO2 += changeNO2;
        currentPM25 += changePM25;
        String message = String.format("Current Air Quality - CO: %.2f ppm, NO2: %.2f ppm, PM2.5: %.2f µg/m³",
                                        currentCO, currentNO2, currentPM25);

        if (currentCO > maxCO || currentNO2 > maxNO2 || currentPM25 > maxPM25) {
            ventilationActive = true;
            message += " | ALERT: Poor Air Quality! Ventilation System Activated.";
        } else {
            ventilationActive = false;
            message += " | Air Quality is Normal. Ventilation System Standby.";
        }
        System.out.println(message);
        log.add(message);
    }

    public void printLog() {
        System.out.println("\nTunnel Air Quality Log:");
        for (String logEntry : log) {
            System.out.println(logEntry);
        }
    }
}

public class c39_TunnelVentilationSystem {
    public static void main(String[] args) {
        TunnelAirQuality airQuality = new TunnelAirQuality(9.0, 1.5, 35.0, 5.0, 0.8, 20.0);
        // // testcase-VT:
        // airQuality.updateAirQuality(59.8, 199.1, 33.1);
    }
}

```

```
//      airQuality.updateAirQuality(-81.9, -146.6, 173.9);
//      airQuality.updateAirQuality(-39.5, 168.9, 61.8);
//      airQuality.updateAirQuality(174.2, -129.6, 59.6);
//      airQuality.updateAirQuality(152.9, 39.0, -35.6);
//      airQuality.updateAirQuality(-157.9, -155.5, -15.5);
//      airQuality.updateAirQuality(154.8, 104.6, -171.0);
//      airQuality.updateAirQuality(-109.7, 44.5, -2.2);
//      airQuality.updateAirQuality(127.2, 180.5, -199.3);
//      airQuality.updateAirQuality(160.3, -52.7, 185.5);
//      airQuality.updateAirQuality(-182.9, 125.8, -90.5);
//      airQuality.updateAirQuality(66.6, 48.7, -183.5);
//      airQuality.updateAirQuality(-169.0, 165.0, -59.3);
//      airQuality.updateAirQuality(-175.8, -163.2, 110.6);
//      airQuality.updateAirQuality(-81.0, -146.6, 195.7);
//      airQuality.updateAirQuality(-13.2, 19.7, 16.8);
//      airQuality.updateAirQuality(-181.6, 111.0, 61.1);
//      airQuality.updateAirQuality(178.0, 172.2, 45.1);
//      airQuality.updateAirQuality(-123.1, 56.1, 22.3);
//      airQuality.updateAirQuality(-127.8, 9.3, -64.0);
//      airQuality.updateAirQuality(194.8, -152.6, -67.5);
//      airQuality.updateAirQuality(52.0, -165.5, 97.0);
//      airQuality.updateAirQuality(195.5, -57.9, -68.8);
//      airQuality.updateAirQuality(-148.5, -172.8, 176.2);
//      airQuality.updateAirQuality(108.4, -0.2, 3.3);
//      airQuality.updateAirQuality(-62.9, -149.2, 106.6);
//      airQuality.updateAirQuality(52.9, 22.6, -25.3);
//      airQuality.updateAirQuality(-26.7, -133.3, -61.9);
//      airQuality.updateAirQuality(-187.0, 146.9, -197.6);
//      airQuality.updateAirQuality(131.7, -52.9, -196.5);
//      airQuality.updateAirQuality(-22.5, 114.2, -122.4);
//      airQuality.updateAirQuality(72.7, -30.3, -11.9);
//      airQuality.updateAirQuality(-184.3, -7.0, -33.8);
//      airQuality.updateAirQuality(11.8, 62.8, 120.7);
//      airQuality.updateAirQuality(-48.7, 145.8, -102.1);
//      airQuality.updateAirQuality(-32.0, 2.4, -118.0);
//      airQuality.updateAirQuality(-171.9, -72.0, 30.9);
//      airQuality.updateAirQuality(-161.4, 136.9, -16.2);
//      airQuality.updateAirQuality(-5.3, 92.8, 42.3);
//      airQuality.updateAirQuality(-132.9, 95.0, -25.8);
//      airQuality.updateAirQuality(-127.2, -21.8, 143.8);
//      airQuality.updateAirQuality(141.0, 136.2, -62.6);
//      airQuality.updateAirQuality(71.8, -109.0, 148.1);
//      airQuality.updateAirQuality(196.7, -25.5, -169.4);
//      airQuality.updateAirQuality(91.8, 126.3, -159.2);
```

```
//      airQuality.updateAirQuality(-4.5, 89.1, 40.9);
//      airQuality.updateAirQuality(11.4, -184.3, 24.3);
//      airQuality.updateAirQuality(-57.1, 152.3, 33.8);
//      airQuality.updateAirQuality(196.5, -199.0, 113.6);
//      airQuality.updateAirQuality(47.6, 69.5, 132.2);
//      airQuality.updateAirQuality(-114.3, -58.3, 154.6);
//      airQuality.updateAirQuality(-17.5, -48.2, -67.8);
//      airQuality.updateAirQuality(133.7, 35.1, 162.4);
//      airQuality.updateAirQuality(-193.2, -74.1, -184.6);
//      airQuality.updateAirQuality(85.4, -99.1, -187.4);
//      airQuality.updateAirQuality(146.0, 26.6, -111.0);
//      airQuality.updateAirQuality(71.3, -147.4, -146.2);
//      airQuality.updateAirQuality(27.2, 65.1, -31.3);
//      airQuality.updateAirQuality(-112.9, 0.9, -8.9);
//      airQuality.updateAirQuality(-28.8, -68.9, 13.9);
//      airQuality.updateAirQuality(2.7, 144.9, -162.5);
//      airQuality.updateAirQuality(43.5, -88.6, 96.8);
//      airQuality.updateAirQuality(-117.8, 12.5, -51.5);
//      airQuality.updateAirQuality(-161.7, -113.5, -25.6);
//      airQuality.updateAirQuality(159.0, -183.5, -199.5);
//      airQuality.updateAirQuality(-86.0, 125.5, 84.9);
//      airQuality.updateAirQuality(-120.8, -2.4, 36.2);
//      airQuality.updateAirQuality(117.5, 135.5, -106.3);
//      airQuality.updateAirQuality(-41.3, -190.5, -21.3);
//      airQuality.updateAirQuality(-123.9, -52.7, 143.2);
//      airQuality.updateAirQuality(47.6, -9.3, 78.4);
//      airQuality.updateAirQuality(113.7, 73.1, -35.1);
//      airQuality.updateAirQuality(66.3, 65.1, 82.7);
//      airQuality.updateAirQuality(-106.4, -66.3, -58.9);
//      airQuality.updateAirQuality(44.9, 107.5, 143.6);
//      airQuality.updateAirQuality(12.0, -126.1, 2.2);
//      airQuality.updateAirQuality(113.9, 41.1, -28.1);
//      airQuality.updateAirQuality(-102.4, -149.6, 162.2);
//      airQuality.updateAirQuality(163.5, -124.4, 0.7);
//      airQuality.updateAirQuality(52.2, -31.2, -100.2);
//      airQuality.updateAirQuality(-176.4, -186.9, -124.4);
//      airQuality.updateAirQuality(-98.4, 17.8, 127.0);
//      airQuality.updateAirQuality(56.1, 94.4, 9.9);
//      airQuality.updateAirQuality(152.4, -181.1, 180.1);
//      airQuality.updateAirQuality(37.7, 13.8, -52.7);
//      airQuality.updateAirQuality(176.3, -195.0, -163.2);
//      airQuality.updateAirQuality(-43.9, -171.2, -41.8);
//      airQuality.updateAirQuality(-66.1, -113.0, -76.4);
//      airQuality.updateAirQuality(61.5, -3.1, 131.7);
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//      airQuality.updateAirQuality(-149.9, -59.7, -87.1);
//      airQuality.updateAirQuality(-50.3, -150.1, 118.0);
//      airQuality.updateAirQuality(23.7, -137.3, 172.4);
//      airQuality.updateAirQuality(-164.7, -77.6, -81.8);
//      airQuality.updateAirQuality(-176.9, 192.2, 158.7);
//      airQuality.updateAirQuality(-74.5, -190.2, -119.2);
//      airQuality.updateAirQuality(150.2, -187.3, -188.4);
//      airQuality.updateAirQuality(-138.2, -168.1, -132.6);
//      airQuality.updateAirQuality(28.6, -147.3, 164.0);
//      airQuality.updateAirQuality(25.3, -178.9, 55.8);
//      airQuality.updateAirQuality(114.8, -124.1, -147.3);

//      // testcase-FT:
//      airQuality.updateAirQuality(-1.7976931348623157E308,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(-1.7976931348623157E308,0.0, 0.0);
//      airQuality.updateAirQuality(-1.7976931348623157E308,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(-1.7976931348623157E308,0.0, 0.0);
//      airQuality.updateAirQuality(-1.7976931348623157E308,0.0, 0.0);
//      airQuality.updateAirQuality(-1.7976931348623157E308,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,1.6707971488720346E308, 1.6707
971488720346E308);
//      airQuality.updateAirQuality(0.0,1.6707971488720346E308, 1.6707
971486627556E308);
//      airQuality.updateAirQuality(0.0,1.6707971488720346E308, 1.6707
971486627556E308);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(-1.6143989328763541E308, -
1.6143989328763541E308, 0.0);
//      airQuality.updateAirQuality(-1.6143989328763541E308, -
1.7976931348623157E308, 0.0);

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//      airQuality.updateAirQuality(-1.6143989328763541E308, -
1.7976931348623157E308, 0.0);
//      airQuality.updateAirQuality(-1.6143989328763541E308, -
1.7976931348623157E308, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(-1.7976931348623157E308,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(-1.7976931348623157E308,0.0, 0.0);
//      airQuality.updateAirQuality(1.7976931348623157E308,1.797693134
8623157E308, 1.7976931348623157E308);
//      airQuality.updateAirQuality(1.7976931348623157E308,1.797693134
8623157E308, 1.7976931348623157E308);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(-1.7976931348623157E308,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(-2.2250738585072014E-
308,0.0, 0.0);
//      airQuality.updateAirQuality(-2.2250738585072014E-
308,0.0, 0.0);
//      airQuality.updateAirQuality(-1.7976931348623157E308,0.0, 0.0);
//      airQuality.updateAirQuality(-1.7976931348623157E308,0.0, 0.0);
//      airQuality.updateAirQuality(-1.7976931348623157E308,0.0, 0.0);
//      airQuality.updateAirQuality(-1.7976931348623157E308,0.0, 0.0);
//      airQuality.updateAirQuality(-1.7976931348623157E308,0.0, 0.0);
//      airQuality.updateAirQuality(-1.7976931348623157E308,0.0, 0.0);
//      airQuality.updateAirQuality(0.0, -1.7976931348623157E308, 0.0);
//      airQuality.updateAirQuality(0.0, -1.7976931348623157E308, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(-2.2250738585072014E-
308,0.0, 0.0);
//      airQuality.updateAirQuality(-0.0,0.0, 0.0);
//      airQuality.updateAirQuality(-0.0,0.0, 0.0);
//      airQuality.updateAirQuality(-0.0,0.0, 0.0);
//      airQuality.updateAirQuality(-0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
```

```
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(-1.7976931348623157E308,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(-0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0, -9.164710099298088E306, -
9.164710099298088E306);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(-1.7976931348623157E308,0.0, 0.0);
//      airQuality.updateAirQuality(-1.7976931348623157E308,0.0, 0.0);
//      airQuality.updateAirQuality(-1.7976931348623157E308,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(-1.7976931348623157E308,0.0, 0.0);
//      airQuality.updateAirQuality(1.7976931348623157E308,1.797693134
8623157E308, 1.7976931348623157E308);
//      airQuality.updateAirQuality(1.7976931348623157E308,1.797693134
8623157E308, 1.7976931348623157E308);
//      airQuality.updateAirQuality(1.7976931348623157E308,1.797693134
8623157E308, 1.7976931348623157E308);
//      airQuality.updateAirQuality(-1.7976931345898426E308,0.0, 0.0);
//      airQuality.updateAirQuality(-6.274302084182735E307, -
1.5580007168806736E308, -1.5580007168806736E308);
//      airQuality.updateAirQuality(-6.274302084182735E307, -
1.5580007168806736E308, -1.5580007168806736E308);
//      airQuality.updateAirQuality(-6.274302084182735E307, -
1.5580007168806736E308, -1.5580007168805715E308);
//      airQuality.updateAirQuality(-6.274302084182735E307, -
1.5580007168806736E308, -1.5580007168805715E308);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(-1.7976931348623157E308,0.0, 0.0);
//      airQuality.updateAirQuality(-1.7976931348623157E308,0.0, 0.0);
//      airQuality.updateAirQuality(-1.7976931348623157E308,0.0, 0.0);
//      airQuality.updateAirQuality(-1.7976931348623157E308,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
//      airQuality.updateAirQuality(0.0,0.0, 0.0);
```

```
//      airQuality.updateAirQuality(-1.7976931348623157E308,0.0, 0.0);
//      airQuality.updateAirQuality(-1.7976931348623157E308,0.0, 0.0);

//      // testcase-VT:
//      airQuality.updateAirQuality(59.8, 199.1, 33.1);
//      airQuality.updateAirQuality(-81.9, -146.6, 173.9);
//      airQuality.updateAirQuality(-39.5, 168.9, 61.8);
//      airQuality.updateAirQuality(174.2, -129.6, 59.6);
//      airQuality.updateAirQuality(152.9, 39.0, -35.6);
//      airQuality.updateAirQuality(-157.9, -155.5, -15.5);
//      airQuality.updateAirQuality(154.8, 104.6, -171.0);
//      airQuality.updateAirQuality(-109.7, 44.5, -2.2);
//      airQuality.updateAirQuality(127.2, 180.5, -199.3);
//      airQuality.updateAirQuality(160.3, -52.7, 185.5);
//      airQuality.updateAirQuality(-182.9, 125.8, -90.5);
//      airQuality.updateAirQuality(66.6, 48.7, -183.5);
//      airQuality.updateAirQuality(-169.0, 165.0, -59.3);
//      airQuality.updateAirQuality(-175.8, -163.2, 110.6);
//      airQuality.updateAirQuality(-81.0, -146.6, 195.7);
//      airQuality.updateAirQuality(-13.2, 19.7, 16.8);
//      airQuality.updateAirQuality(-181.6, 111.0, 61.1);
//      airQuality.updateAirQuality(178.0, 172.2, 45.1);
//      airQuality.updateAirQuality(-123.1, 56.1, 22.3);
//      airQuality.updateAirQuality(-127.8, 9.3, -64.0);
//      airQuality.updateAirQuality(194.8, -152.6, -67.5);
//      airQuality.updateAirQuality(52.0, -165.5, 97.0);
//      airQuality.updateAirQuality(195.5, -57.9, -68.8);
//      airQuality.updateAirQuality(-148.5, -172.8, 176.2);
//      airQuality.updateAirQuality(108.4, -0.2, 3.3);
//      airQuality.updateAirQuality(-62.9, -149.2, 106.6);
//      airQuality.updateAirQuality(52.9, 22.6, -25.3);
//      airQuality.updateAirQuality(-26.7, -133.3, -61.9);
//      airQuality.updateAirQuality(-187.0, 146.9, -197.6);
//      airQuality.updateAirQuality(131.7, -52.9, -196.5);
//      airQuality.updateAirQuality(-22.5, 114.2, -122.4);
//      airQuality.updateAirQuality(72.7, -30.3, -11.9);
//      airQuality.updateAirQuality(-184.3, -7.0, -33.8);
//      airQuality.updateAirQuality(11.8, 62.8, 120.7);
//      airQuality.updateAirQuality(-48.7, 145.8, -102.1);
//      airQuality.updateAirQuality(-32.0, 2.4, -118.0);
//      airQuality.updateAirQuality(-171.9, -72.0, 30.9);
//      airQuality.updateAirQuality(-161.4, 136.9, -16.2);
//      airQuality.updateAirQuality(-5.3, 92.8, 42.3);
//      airQuality.updateAirQuality(-132.9, 95.0, -25.8);
```

```
//      airQuality.updateAirQuality(-127.2, -21.8, 143.8);
//      airQuality.updateAirQuality(141.0, 136.2, -62.6);
//      airQuality.updateAirQuality(71.8, -109.0, 148.1);
//      airQuality.updateAirQuality(196.7, -25.5, -169.4);
//      airQuality.updateAirQuality(91.8, 126.3, -159.2);
//      airQuality.updateAirQuality(-4.5, 89.1, 40.9);
//      airQuality.updateAirQuality(11.4, -184.3, 24.3);
//      airQuality.updateAirQuality(-57.1, 152.3, 33.8);
//      airQuality.updateAirQuality(196.5, -199.0, 113.6);
//      airQuality.updateAirQuality(47.6, 69.5, 132.2);
//      airQuality.updateAirQuality(-114.3, -58.3, 154.6);
//      airQuality.updateAirQuality(-17.5, -48.2, -67.8);
//      airQuality.updateAirQuality(133.7, 35.1, 162.4);
//      airQuality.updateAirQuality(-193.2, -74.1, -184.6);
//      airQuality.updateAirQuality(85.4, -99.1, -187.4);
//      airQuality.updateAirQuality(146.0, 26.6, -111.0);
//      airQuality.updateAirQuality(71.3, -147.4, -146.2);
//      airQuality.updateAirQuality(27.2, 65.1, -31.3);
//      airQuality.updateAirQuality(-112.9, 0.9, -8.9);
//      airQuality.updateAirQuality(-28.8, -68.9, 13.9);
//      airQuality.updateAirQuality(2.7, 144.9, -162.5);
//      airQuality.updateAirQuality(43.5, -88.6, 96.8);
//      airQuality.updateAirQuality(-117.8, 12.5, -51.5);
//      airQuality.updateAirQuality(-161.7, -113.5, -25.6);
//      airQuality.updateAirQuality(159.0, -183.5, -199.5);
//      airQuality.updateAirQuality(-86.0, 125.5, 84.9);
//      airQuality.updateAirQuality(-120.8, -2.4, 36.2);
//      airQuality.updateAirQuality(117.5, 135.5, -106.3);
//      airQuality.updateAirQuality(-41.3, -190.5, -21.3);
//      airQuality.updateAirQuality(-123.9, -52.7, 143.2);
//      airQuality.updateAirQuality(47.6, -9.3, 78.4);
//      airQuality.updateAirQuality(113.7, 73.1, -35.1);
//      airQuality.updateAirQuality(66.3, 65.1, 82.7);
//      airQuality.updateAirQuality(-106.4, -66.3, -58.9);
//      airQuality.updateAirQuality(44.9, 107.5, 143.6);
//      airQuality.updateAirQuality(12.0, -126.1, 2.2);
//      airQuality.updateAirQuality(113.9, 41.1, -28.1);
//      airQuality.updateAirQuality(-102.4, -149.6, 162.2);
//      airQuality.updateAirQuality(163.5, -124.4, 0.7);
//      airQuality.updateAirQuality(52.2, -31.2, -100.2);
//      airQuality.updateAirQuality(-176.4, -186.9, -124.4);
//      airQuality.updateAirQuality(-98.4, 17.8, 127.0);
//      airQuality.updateAirQuality(56.1, 94.4, 9.9);
//      airQuality.updateAirQuality(152.4, -181.1, 180.1);
```

```
//      airQuality.updateAirQuality(37.7, 13.8, -52.7);
//      airQuality.updateAirQuality(176.3, -195.0, -163.2);
//      airQuality.updateAirQuality(-43.9, -171.2, -41.8);
//      airQuality.updateAirQuality(-66.1, -113.0, -76.4);
//      airQuality.updateAirQuality(61.5, -3.1, 131.7);
//      airQuality.updateAirQuality(-149.9, -59.7, -87.1);
//      airQuality.updateAirQuality(-50.3, -150.1, 118.0);
//      airQuality.updateAirQuality(23.7, -137.3, 172.4);
//      airQuality.updateAirQuality(-164.7, -77.6, -81.8);
//      airQuality.updateAirQuality(-176.9, 192.2, 158.7);
//      airQuality.updateAirQuality(-74.5, -190.2, -119.2);
//      airQuality.updateAirQuality(150.2, -187.3, -188.4);
//      airQuality.updateAirQuality(-138.2, -168.1, -132.6);
//      airQuality.updateAirQuality(28.6, -147.3, 164.0);
//      airQuality.updateAirQuality(25.3, -178.9, 55.8);
//      airQuality.updateAirQuality(114.8, -124.1, -147.3);
```

```
// testcase-UVT:
```

```
airQuality.updateAirQuality(59.8, 199.1, 33.1);
airQuality.updateAirQuality(-81.9, -146.6, 173.9);
airQuality.updateAirQuality(-39.5, 168.9, 61.8);
airQuality.updateAirQuality(174.2, -129.6, 59.6);
airQuality.updateAirQuality(152.9, 39.0, -35.6);
airQuality.updateAirQuality(-157.9, -155.5, -15.5);
airQuality.updateAirQuality(154.8, 104.6, -171.0);
airQuality.updateAirQuality(-109.7, 44.5, -2.2);
airQuality.updateAirQuality(127.2, 180.5, -199.3);
airQuality.updateAirQuality(160.3, -52.7, 185.5);
airQuality.updateAirQuality(-182.9, 125.8, -90.5);
airQuality.updateAirQuality(66.6, 48.7, -183.5);
airQuality.updateAirQuality(-169.0, 165.0, -59.3);
airQuality.updateAirQuality(-175.8, -163.2, 110.6);
airQuality.updateAirQuality(-81.0, -146.6, 195.7);
airQuality.updateAirQuality(-13.2, 19.7, 16.8);
airQuality.updateAirQuality(-181.6, 111.0, 61.1);
airQuality.updateAirQuality(178.0, 172.2, 45.1);
airQuality.updateAirQuality(-123.1, 56.1, 22.3);
airQuality.updateAirQuality(-127.8, 9.3, -64.0);
airQuality.updateAirQuality(194.8, -152.6, -67.5);
airQuality.updateAirQuality(52.0, -165.5, 97.0);
airQuality.updateAirQuality(195.5, -57.9, -68.8);
airQuality.updateAirQuality(-148.5, -172.8, 176.2);
airQuality.updateAirQuality(108.4, -0.2, 3.3);
airQuality.updateAirQuality(-62.9, -149.2, 106.6);
```

```
airQuality.updateAirQuality(52.9, 22.6, -25.3);
airQuality.updateAirQuality(-26.7, -133.3, -61.9);
airQuality.updateAirQuality(-187.0, 146.9, -197.6);
airQuality.updateAirQuality(131.7, -52.9, -196.5);
airQuality.updateAirQuality(-22.5, 114.2, -122.4);
airQuality.updateAirQuality(72.7, -30.3, -11.9);
airQuality.updateAirQuality(-184.3, -7.0, -33.8);
airQuality.updateAirQuality(11.8, 62.8, 120.7);
airQuality.updateAirQuality(-48.7, 145.8, -102.1);
airQuality.updateAirQuality(-32.0, 2.4, -118.0);
airQuality.updateAirQuality(-171.9, -72.0, 30.9);
airQuality.updateAirQuality(-161.4, 136.9, -16.2);
airQuality.updateAirQuality(-5.3, 92.8, 42.3);
airQuality.updateAirQuality(-132.9, 95.0, -25.8);
airQuality.updateAirQuality(-127.2, -21.8, 143.8);
airQuality.updateAirQuality(141.0, 136.2, -62.6);
airQuality.updateAirQuality(71.8, -109.0, 148.1);
airQuality.updateAirQuality(196.7, -25.5, -169.4);
airQuality.updateAirQuality(91.8, 126.3, -159.2);
airQuality.updateAirQuality(-4.5, 89.1, 40.9);
airQuality.updateAirQuality(11.4, -184.3, 24.3);
airQuality.updateAirQuality(-57.1, 152.3, 33.8);
airQuality.updateAirQuality(196.5, -199.0, 113.6);
airQuality.updateAirQuality(47.6, 69.5, 132.2);
airQuality.updateAirQuality(-114.3, -58.3, 154.6);
airQuality.updateAirQuality(-17.5, -48.2, -67.8);
airQuality.updateAirQuality(133.7, 35.1, 162.4);
airQuality.updateAirQuality(-193.2, -74.1, -184.6);
airQuality.updateAirQuality(85.4, -99.1, -187.4);
airQuality.updateAirQuality(146.0, 26.6, -111.0);
airQuality.updateAirQuality(71.3, -147.4, -146.2);
airQuality.updateAirQuality(27.2, 65.1, -31.3);
airQuality.updateAirQuality(-112.9, 0.9, -8.9);
airQuality.updateAirQuality(-28.8, -68.9, 13.9);
airQuality.updateAirQuality(2.7, 144.9, -162.5);
airQuality.updateAirQuality(43.5, -88.6, 96.8);
airQuality.updateAirQuality(-117.8, 12.5, -51.5);
airQuality.updateAirQuality(-161.7, -113.5, -25.6);
airQuality.updateAirQuality(159.0, -183.5, -199.5);
airQuality.updateAirQuality(-86.0, 125.5, 84.9);
airQuality.updateAirQuality(-120.8, -2.4, 36.2);
airQuality.updateAirQuality(117.5, 135.5, -106.3);
airQuality.updateAirQuality(-41.3, -190.5, -21.3);
airQuality.updateAirQuality(-123.9, -52.7, 143.2);
```

```

        airQuality.updateAirQuality(47.6, -9.3, 78.4);
        airQuality.updateAirQuality(113.7, 73.1, -35.1);
        airQuality.updateAirQuality(66.3, 65.1, 82.7);
        airQuality.updateAirQuality(-106.4, -66.3, -58.9);
        airQuality.updateAirQuality(44.9, 107.5, 143.6);
        airQuality.updateAirQuality(12.0, -126.1, 2.2);
        airQuality.updateAirQuality(113.9, 41.1, -28.1);
        airQuality.updateAirQuality(-102.4, -149.6, 162.2);
        airQuality.updateAirQuality(163.5, -124.4, 0.7);
        airQuality.updateAirQuality(52.2, -31.2, -100.2);
        airQuality.updateAirQuality(-176.4, -186.9, -124.4);
        airQuality.updateAirQuality(-98.4, 17.8, 127.0);
        airQuality.updateAirQuality(56.1, 94.4, 9.9);
        airQuality.updateAirQuality(152.4, -181.1, 180.1);
        airQuality.updateAirQuality(37.7, 13.8, -52.7);
        airQuality.updateAirQuality(176.3, -195.0, -163.2);
        airQuality.updateAirQuality(-43.9, -171.2, -41.8);
        airQuality.updateAirQuality(-66.1, -113.0, -76.4);
        airQuality.updateAirQuality(61.5, -3.1, 131.7);
        airQuality.updateAirQuality(-149.9, -59.7, -87.1);
        airQuality.updateAirQuality(-50.3, -150.1, 118.0);
        airQuality.updateAirQuality(23.7, -137.3, 172.4);
        airQuality.updateAirQuality(-164.7, -77.6, -81.8);
        airQuality.updateAirQuality(-176.9, 192.2, 158.7);
        airQuality.updateAirQuality(-74.5, -190.2, -119.2);
        airQuality.updateAirQuality(150.2, -187.3, -188.4);
        airQuality.updateAirQuality(-138.2, -168.1, -132.6);
        airQuality.updateAirQuality(28.6, -147.3, 164.0);
        airQuality.updateAirQuality(25.3, -178.9, 55.8);
        airQuality.updateAirQuality(114.8, -124.1, -147.3);
    //      airQuality.printLog();
    }
}

```

- Coverage report

1. Test case number: 100
2. V-Method testing branch coverage: 75%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.7 %	6	836	842
✓ src	0.7 %	6	836	842
✓ code	0.7 %	6	836	842
✓ c39_TunnelVentilationSystem.java	75.0 %	6	2	8
> TunnelAirQuality	75.0 %	6	2	8
> c39_TunnelVentilationSystem		0	0	0

3. Vibration testing instruction coverage: 96.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	3.0 %	621	20,039	20,660
✓ src	3.0 %	621	20,039	20,660
✓ code	3.0 %	621	20,039	20,660
✓ c39_TunnelVentilationSystem.java	96.6 %	621	22	643
> TunnelAirQuality	85.3 %	110	19	129
> c39_TunnelVentilationSystem	99.4 %	511	3	514

4. Fuzz testing branch coverage: 50%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.5 %	4	838	842
✓ src	0.5 %	4	838	842
✓ code	0.5 %	4	838	842
✓ c39_TunnelVentilationSystem.java	50.0 %	4	4	8
> TunnelAirQuality	50.0 %	4	4	8
> c39_TunnelVentilationSystem		0	0	0

5. Fuzz testing instruction coverage: 96.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	3.0 %	621	20,018	20,639
✓ src	3.0 %	621	20,018	20,639
✓ code	3.0 %	621	20,018	20,639
✓ c39_TunnelVentilationSystem.java	96.6 %	621	22	643
> TunnelAirQuality	85.3 %	110	19	129
> c39_TunnelVentilationSystem	99.4 %	511	3	514

6. Z3 branch coverage: 75%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.7 %	6	834	840
✓ src	0.7 %	6	834	840
✓ code	0.7 %	6	834	840
✓ c39_TunnelVentilationSystem.java	75.0 %	6	2	8
> TunnelAirQuality	75.0 %	6	2	8
> c39_TunnelVentilationSystem		0	0	0

7. Z3 instruction coverage: 96.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	2.8 %	621	21,761	22,382
✓ src	2.8 %	621	21,761	22,382
✓ code	2.8 %	621	21,761	22,382
✓ c39_TunnelVentilationSystem.java	96.6 %	621	22	643
> TunnelAirQuality	85.3 %	110	19	129
> c39_TunnelVentilationSystem	99.4 %	511	3	514

8. Unimproved V-method branch coverage: 75%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.7 %	6	834	840
✓ src	0.7 %	6	834	840
✓ code	0.7 %	6	834	840
✓ c39_TunnelVentilationSystem.java	75.0 %	6	2	8
> TunnelAirQuality	75.0 %	6	2	8
> c39_TunnelVentilationSystem		0	0	0

9. Unimproved V-method instruction coverage: 96.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	2.8 %	621	21,761	22,382
✓ src	2.8 %	621	21,761	22,382
✓ code	2.8 %	621	21,761	22,382
✓ c39_TunnelVentilationSystem.java	96.6 %	621	22	643
> TunnelAirQuality	85.3 %	110	19	129
> c39_TunnelVentilationSystem	99.4 %	511	3	514